

Caleb Kornegay

Aspiring Software Engineer with an Industry Internship, Teaching Experience, and Projects in Systems, Web, and Embedded Development

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EXPERIENCE

Siemens Healthineers, Knoxville, TN — *Electrical Research and Development Intern*

January 2024 - August 2024

Developed cross-platform GUIs in PySide6 and WPF for sensor and board interfaces, improving team testability

Designed and implemented Arduino-based distributed system for real-time sensor data collection and processing

Conducted CUDA/C++ research to improve data inflow processing and decrease CPU overhead

Built a REST API on Zephyr RTOS and integrated frontend in React for live data display and board state augmentation

University of Tennessee - Knoxville — *Teacher's Assistant*

August 2022 - December 2023

Held regular office hours to support 100+ students in understanding core algorithms, memory management, and data structures

EDUCATION

University of Tennessee - Knoxville — *B.S. Computer Science*

August 2021 - May 2023

GPA: 3.99/4.0, Summa Cum Laude

PROJECTS

OpenTicket — *Go, React, React Native, SQL, Solidity*

Developed full-stack decentralized ticketing platform with React web frontend, React Native mobile apps, Go REST API, and Solidity smart contracts for on-chain ticket issuance and ownership tracking to combat overpriced ticket fees

Deployed backend on AWS using CDK utilizing Lambda with VPC-secured RDS database, enabling scalable, serverless infrastructure

Fzk — *Rust*

Developed a cross-platform command-line tool for interactively searching and terminating processes using fuzzy matching, enhancing system administration efficiency, with plans for future enhancements such as process sorting and multi-process termination for ease-of-use

SKILLS

Serial protocols - UART, SPI, I2C, Modbus

Hardware debugging - JTAG, oscilloscope, ADC/DAC

AWS - Lambda, RDS, VPC

Scripting - Bash and Python

Tools: Git, CMake, AWS CDK, CI/CD, Zephyr RTOS, PostgreSQL, Cargo

AWARDS

Outstanding Computer Science Sophomore 2023

LANGUAGES

C/C++

Typescript/HTML/CSS

C#

SQL

Rust

Go